

Program : Diploma in Architecture	
Course Code : 6181C	Course Title: Repairs and maintenance of structures
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To learn about types of maintenance techniques
- To understand causes of various types of damages.
- To know about relevant materials for repair.
- To learn methods of retrofitting for different structures.

Course Prerequisites:

Topic	Course code	Course name	Semester
Building Construction		Building construction and Materials	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Explain causes of damages and decide the type of maintenance required for a given damaged structure	14	Understanding
CO2	Determine the relevant method for repair of damage	14	Understanding
CO3	Apply relevant method of retrofitting for re-strengthening of masonry structures	15	Understanding
CO4	Determine relevant technique to restore the damages of the given structural elements.	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2	2					2
CO3	2	2					
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain causes of damages and decide the type of maintenance required for a given damaged structure		
M1.01	Recognize the influence of environmental elements	4	Understanding
M1.02	Explain different types of Maintenances and its need	3	Understanding
M1.03	Identify the type of damage	3	Understanding
M1.04	Decide the type of maintenance required	4	Understanding
Contents: Life expectancy of different types of buildings, influence of environmental elements: heat, moisture, precipitation & frost on buildings, effect of biological agents like fungus, moss, plants, trees, algae, termite control & prevention, chemical attack on building materials & components, aspects of fire & fire prevention on buildings, impact of pollution on buildings. Types of Maintenances - repair, retrofitting, re-strengthening, rehabilitation and restoration. - Necessity, objectives and importance of maintenance. - Pre - and post - monsoon maintenance. - Various aspects of visual observations for detection of damages. Causes of damages due to distress, earthquake, wind, flood, dampness, corrosion, fire, deterioration, termites, pollution and foundation settlement.			
CO2	Determine the relevant method for repair of damage		
M2.01	Select materials for repairs	4	Understanding
M2.02	Identify essential parameters for maintenance	4	Understanding
M2.03	Explain water proofing techniques	4	Understanding

M2.04	Describe repairing methods for RCC	3	Understanding
	Series Test – I	1	
Contents: Building Repairs Materials for repair: special mortar & concrete, chemicals, special cements & high-grade concrete, admixtures, techniques for repair, Surface repair: material selection, surface preparation, rust eliminators & polymers coating, Repair of cracks in concrete: methods of repair, epoxy injection, mortar repair for cracks: guniting & shotcreting, Waterproofing of concrete roofs. Repairing materials for RCC: epoxy resins, epoxy mortar, cement mortar impregnated with polypropylene, silicon, polymer concrete composites, sealants, fiber reinforcement concrete, emulsions and paints			
CO3	Apply relevant method of retrofitting for re-strengthening of masonry structures		
M3.01	Identify the type of damage	5	Understanding
M3.02	Select the repair method for fixing damage	5	Applying
M3.03	Explain the method of application	5	Understanding
Contents: Common defects in buildings - Building failures- Causes and effects - Cracks in buildings: types, classification, investigation Causes of cracks in walls - bulging of wall, shrinkage, bonding, shear, tensile, vegetation. - Probable crack location: junction of main and cross wall, junction of slab and wall, cracks in masonry joints. - Repair methods based on crack type - For minor & medium cracks (width 0.5 mm to 5mm): grouting and for major cracks (width more than 5mm): fixing mesh across cracks, RCC band, installing ferro-cement plates at corners, dowel bars, propping of load bearing. - Remedial measures for dampness & efflorescence in wall.			
CO4	Determine relevant technique to restore the damages of the given structural elements		
M4.01	Describe strengthening methods for live cracks	4	Understanding
M4.02	Identify building cracks and its prevention	4	Understanding
M4.03	Identify strengthening methods for live cracks	3	Understanding
M4.04	Examine the repair stages and select repair options	4	Applying
	Series Test – II	1	

Contents:

Repair stages such as concrete removal and surface preparation, fixing suitable formwork, bonding/passive coat and repair application, various methods of surface preparation. Repair options such as grouting, patch repairs, carbonated concrete, cleaning the corroded steel.

Fundamental Strengthening measures: beam strengthening, column strengthening, shoring, under pinning and jacketing.

-Strengthening methods for live cracks such as addition of reinforcements, Jacketing, brackets, collars, supplementary members i.e. shoring, underpinning and propping of framed structure concrete overlays, corrosion protection such as jacketing. - Building cracks and its prevention, common methods for dormant crack repairs such as Epoxy injection, grooving and sealing, stitching, grouting and guniting/ shotcreting.

Text / Reference:

T/R	Book Title/Author
T1	Gahlot, P. S., Sharma, S., Building Repair and Maintenance Management, CBS Publishers & Distributors Pvt. Ltd., New Delhi
R1	Guha, P. K., Maintenance and Repairs of Buildings, New Central Book Agencies
R2	Hutchin Son, B. D., Maintenance and Repairs of Buildings, Newnes-Butterworth
R3	Relevant BIS codes